wfmR Article Demo

Typography, colour, one wide table, and chart metadata

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Overview

This article demonstrates the shared HTML and PDF defaults in `wfmR`. The HTML render uses the default system sans-serif stack, while the PDF render uses the LaTeX defaults from the package extension.

The examples use ggplot2 (Wickham 2016) and the built-in `mtcars`, `AirPassengers`, and `volcano` datasets, so the demo can render anywhere the package is installed. The grammar-of-graphics framing follows Wilkinson (2005), while the prose uses ordinary Quarto markdown. Generic coloured labels such as salmon, teal, purple, and lime inherit the note highlight style.

Use `mainfont` under `wfmr-html` only when a paper-like HTML note should read closer to the PDF output. Leave it unset for the default sans-serif article style.

The report style is intentionally more editorial than the academic style. Chart titles and subtitles sit above the image, notes and sources sit below, and the surrounding page uses the soft Rose Pine-inspired palette for links, labels, tables, and callout accents.

This expanded demo deliberately references each chart and the single table from nearby prose. In the HTML render those references become hoverable links. In the PDF render they provide a useful check that floats remain close to the paragraph that introduces them.

Palette

The primary palette is ordered for categorical charts: salmon, teal, purple, lime, yellow, blue, and pink. Each base colour also has named tints from 90 to 10 percent of the original colour. The palette is intentionally softer than publication-first categorical schemes, but it still follows the general principle that colour choices should be documented rather than left implicit (Brewer, Hatchard, and Harrower 2003).

Chart 1 uses every default colour. It is a simple diagnostic for checking the categorical order in HTML, PDF, and revealjs output. It also gives the PDF renderer an early figure reference before enough prose has accumulated to force a difficult page break.

Dummy placement text follows the first reference so the PDF renderer has a choice to make. The text is deliberately content-light: it exists to fill the page between the cross-reference and the float, letting the chart move to the bottom of the current page or the top of the next page. In normal analysis this space would describe the sample, the measurement choices, and the reason the diagnostic appears before the substantive plots.

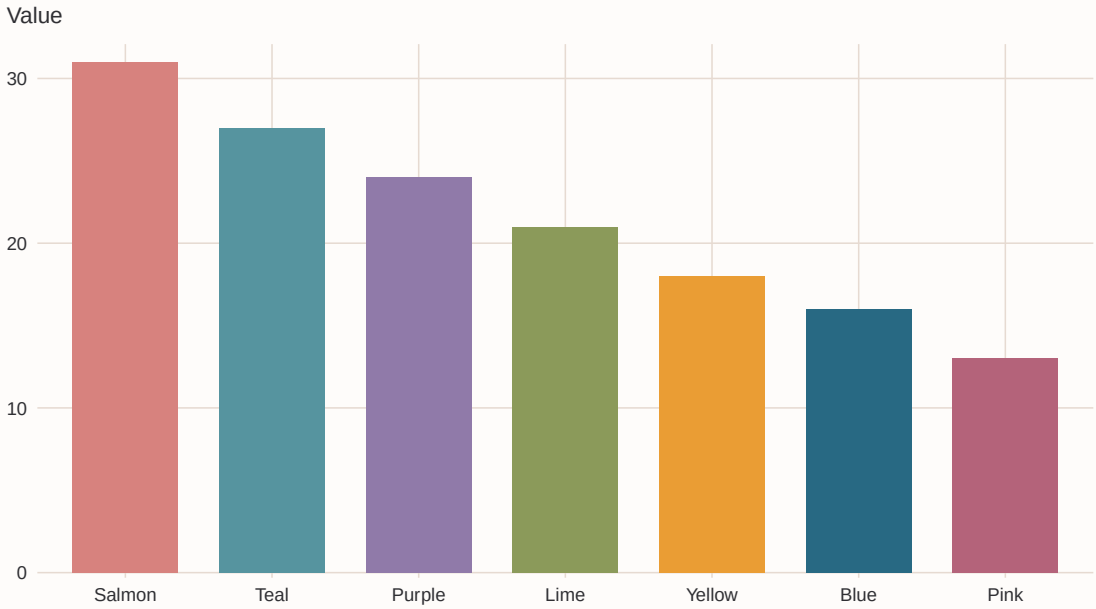
Chart Metadata

The chart metadata hook lets chunks use `chart-title`, `chart-subtitle`, `chart-notes`, and `chart-source` rather than hand-building captions. The rendered caption and notes stay consistent across HTML, PDF, and revealjs. This section uses two chart shapes to check that the metadata works across common report objects. Chart 2 introduces a bivariate pattern, and Chart 3 changes the geometry while keeping the same caption structure.

These objects are not intended as a substantive vehicle analysis. They are small, portable examples that make cross-references, captions, subtitles, notes, sources, and legends visible in a single document. That is useful because visual hierarchy is easiest to judge when the same object appears in ordinary surrounding prose, not only in isolated screenshots. The same logic is behind the classic advice to use graphics as part of the argument rather than as decoration (Cleveland 1993; Tufte 2001).

This paragraph is another placement spacer. It is intentionally mundane and repetitive,

Chart 1: The default palette supports larger categorical charts
Seven bars use the primary wfmR order



Notes: The full seven-colour set is useful for brand consistency; for analytical comparisons, direct labels remain preferable when categories are dense.
Source: Illustrative data.

because the point is to create the conditions under which LaTeX has to decide whether the next figure belongs at the bottom of the current page or the top of the next page. The object should not remain inline merely because it was inserted at this point in the source.

A second spacer paragraph makes the test more obvious in the generated PDF. In a real report it might discuss robustness checks, modelling assumptions, or data cleaning decisions. Here it simply gives the float queue enough surrounding text to reveal whether the theme is using genuine top and bottom floats.

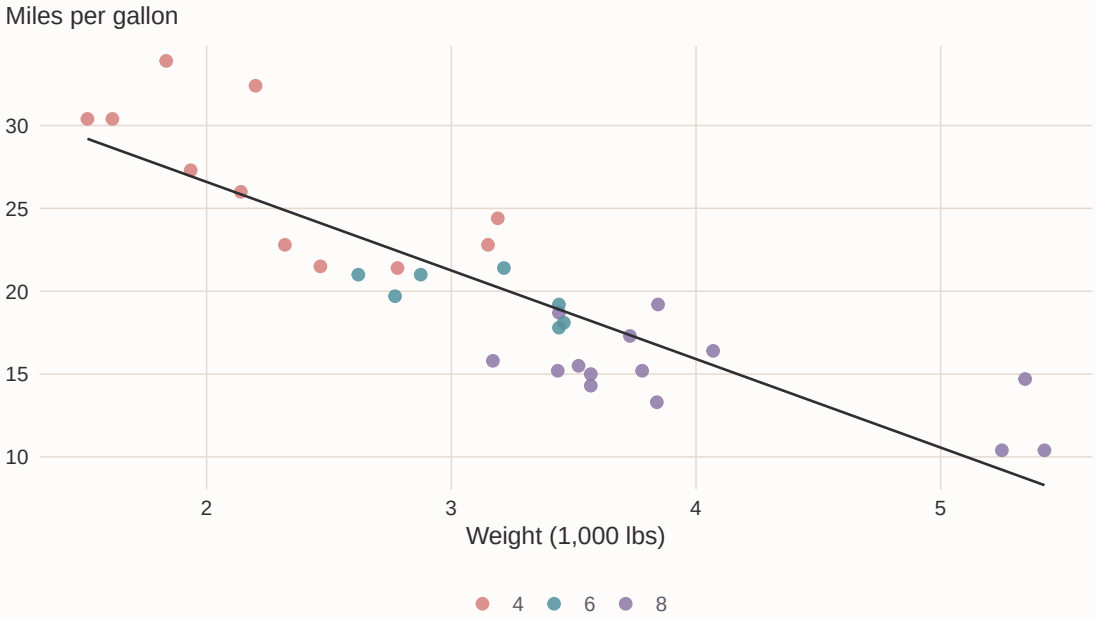
Distribution Checks

The next pair of objects puts a more crowded chart near the only table in the demo. Chart 4 checks whether overlapping fills, legend keys, and axis labels remain quiet enough for report prose. Table 1 is deliberately wide, with enough rows and columns to test HTML table styling and PDF float placement without filling the document with repeated table examples.

The goal is practical rather than statistical. If a theme works only for one bar chart and one tiny summary table, it will not survive normal project work. Distribution plots, a wide table, and nearby narrative text create a better stress test for spacing and cross-reference behaviour.

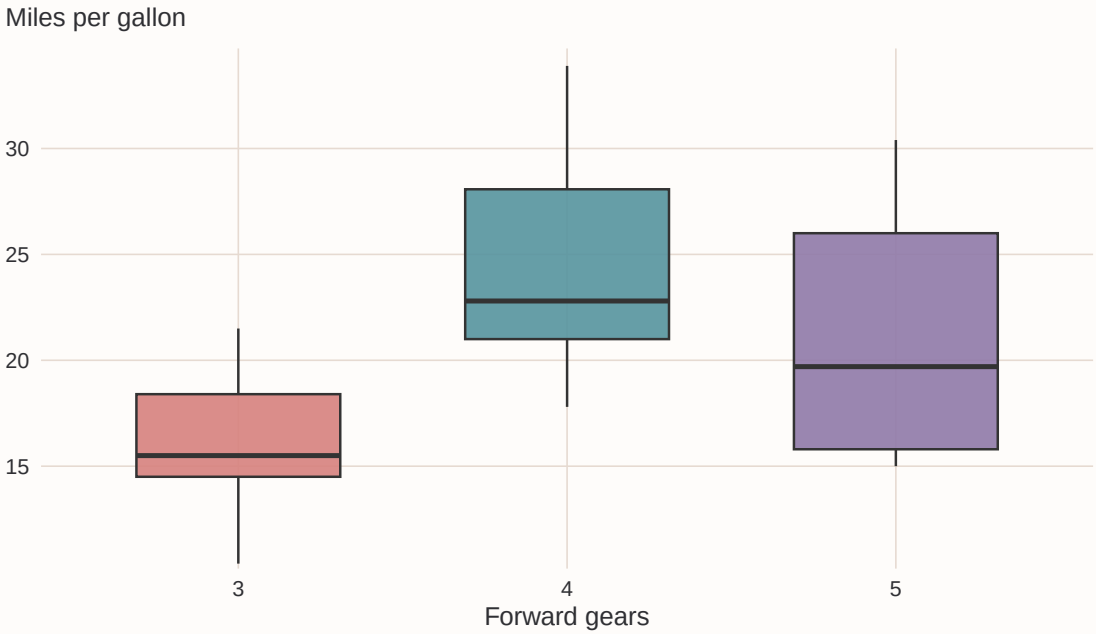
This dummy paragraph is here to push the next figure away from its first reference. The desired behaviour is not inline placement. The desired behaviour is a top or bottom

Chart 2: Fuel economy is lower for heavier cars
A small example using the built-in mtcars dataset



Notes: The fitted line is illustrative. Colours come from the wfmR primary palette.
Source: R datasets package.

Chart 3: Fuel economy also varies by transmission gearing
Boxplots provide a second categorical chart example



Notes: Groups are small because this is the built-in mtcars dataset.
Source: R datasets package.

float that remains close enough to the reference while keeping the surrounding text block readable.

The same stress test applies to the following table. A table emitted as `longtable` cannot float, so the PDF version of this demo now emits a real LaTeX `table` environment. The table is wide enough to exercise scaling in PDF and horizontal breathing room in HTML, while still remaining one grouped object with its caption, notes, and source.

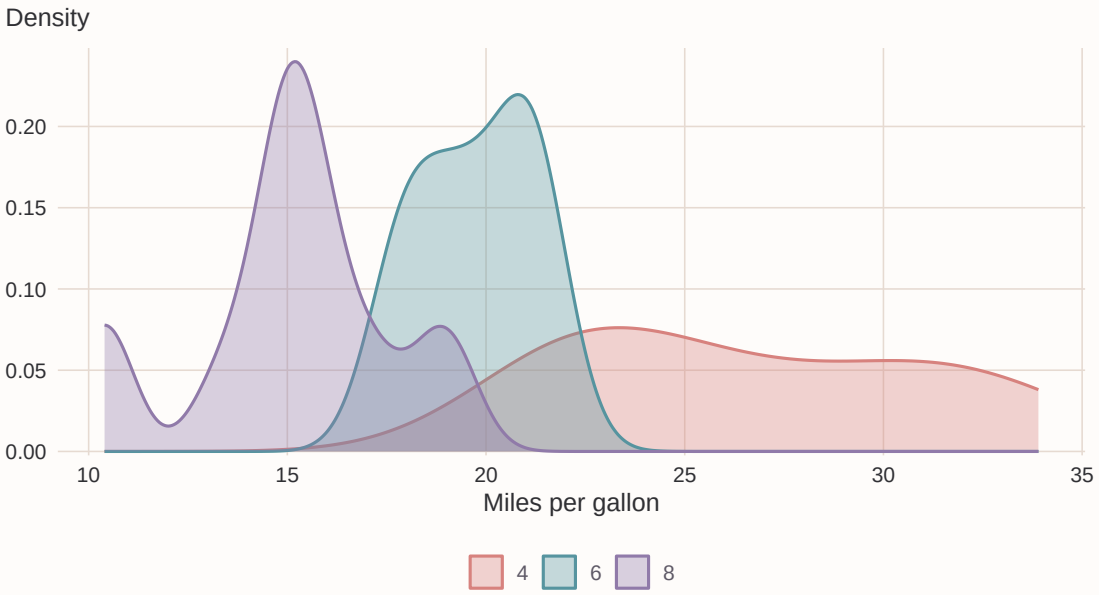
Time Series

The built-in `AirPassengers` series gives the article demo a non-vehicle example and a different visual rhythm. Chart 5 uses the sequentially ordered time index. The example is useful for checking that figure placement remains sensible after the earlier wide table has already entered the float queue.

The line chart also checks that the palette can support a single primary accent without making every graphic look like a categorical palette showcase. This is the report counterpart to a typical short empirical note: a trend, a few paragraphs explaining why the object is present.

The next two dummy paragraphs are deliberately ordinary. They create a page where the time-series figure can plausibly fit at the bottom, but may need to move to the top of the following page if the accumulated caption and notes make that impossible. This is the sort of case where an inline figure would leave an awkward hole in the prose.

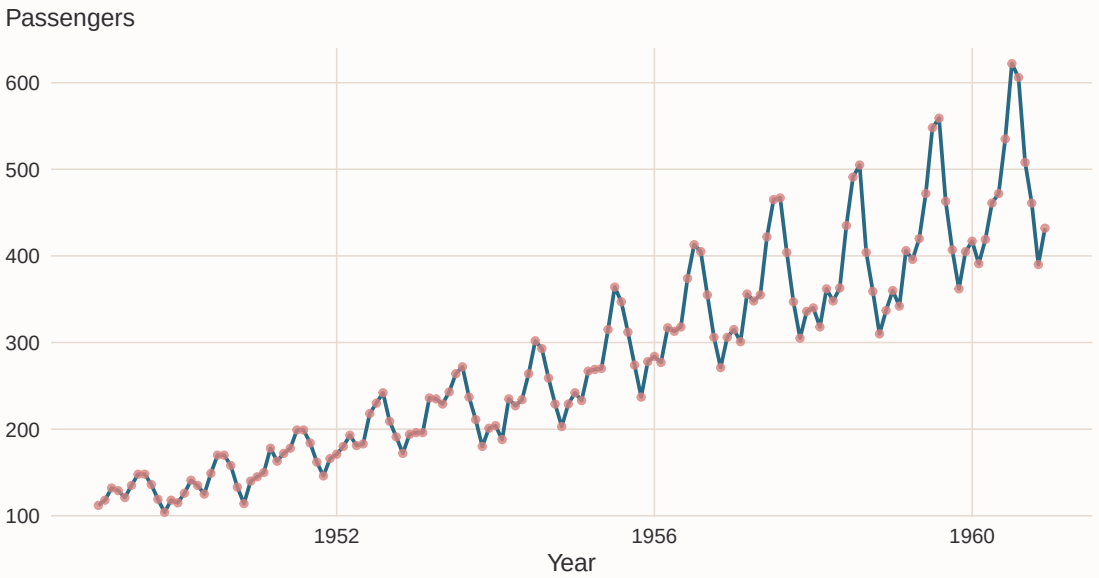
Chart 4: Fuel economy distributions are visibly different by cylinder count
Density curves test alpha blending, legends, and axis spacing



Notes: The curves are illustrative and use the small built-in mtcars data.
Source: R datasets package.

The extra prose remains here to keep testing placement pressure after the table section. The cross-reference should do the navigational work, while LaTeX handles whether the figure sits at the top or bottom of a nearby page.

Chart 5: Passenger counts rise sharply in the late 1950s
Monthly totals from the built-in AirPassengers series



Notes: The example is included to exercise a time-series chart in the same theme.
Source: R datasets package.

Table 1: Wide vehicle summary for table placement

Toyota Corona	4	3	Automatic	21.5	2.46	97	20.01	3.70	120
Datsun 710	4	4	Manual	22.8	2.32	93	18.61	3.85	108
Fiat 128	4	4	Manual	32.4	2.20	66	19.47	4.08	79
Fiat X1-9	4	4	Manual	27.3	1.94	66	18.90	4.08	79
Honda Civic	4	4	Manual	30.4	1.61	52	18.52	4.93	76
Merc 230	4	4	Automatic	22.8	3.15	95	22.90	3.92	141
Merc 240D	4	4	Automatic	24.4	3.19	62	20.00	3.69	147
Toyota Corolla	4	4	Manual	33.9	1.83	65	19.90	4.22	71
Volvo 142E	4	4	Manual	21.4	2.78	109	18.60	4.11	121
Lotus Europa	4	5	Manual	30.4	1.51	113	16.90	3.77	95
Porsche 914-2	4	5	Manual	26.0	2.14	91	16.70	4.43	120
Hornet 4 Drive	6	3	Automatic	21.4	3.21	110	19.44	3.08	258
Valiant	6	3	Automatic	18.1	3.46	105	20.22	2.76	225
Mazda RX4	6	4	Manual	21.0	2.62	110	16.46	3.90	160
Mazda RX4 Wag	6	4	Manual	21.0	2.88	110	17.02	3.90	160
Merc 280	6	4	Automatic	19.2	3.44	123	18.30	3.92	168
Merc 280C	6	4	Automatic	17.8	3.44	123	18.90	3.92	168
Ferrari Dino	6	5	Manual	19.7	2.77	175	15.50	3.62	145
AMC Javelin	8	3	Automatic	15.2	3.44	150	17.30	3.15	304
Cadillac Fleetwood	8	3	Automatic	10.4	5.25	205	17.98	2.93	472
Camaro Z28	8	3	Automatic	13.3	3.84	245	15.41	3.73	350
Chrysler Imperial	8	3	Automatic	14.7	5.34	230	17.42	3.23	440
Dodge Challenger	8	3	Automatic	15.5	3.52	150	16.87	2.76	318
Duster 360	8	3	Automatic	14.3	3.57	245	15.84	3.21	360
Hornet Sportabout	8	3	Automatic	18.7	3.44	175	17.02	3.15	360
Lincoln Continental	8	3	Automatic	10.4	5.42	215	17.82	3.00	460
Merc 450SE	8	3	Automatic	16.4	4.07	180	17.40	3.07	276
Merc 450SL	8	3	Automatic	17.3	3.73	180	17.60	3.07	276
Merc 450SLC	8	3	Automatic	15.2	3.78	180	18.00	3.07	276
Pontiac Firebird	8	3	Automatic	19.2	3.85	175	17.05	3.08	400
Ford Pantera L	8	5	Manual	15.8	3.17	264	14.50	4.22	351
Maserati Bora	8	5	Manual	15.0	3.57	335	14.60	3.54	301

Notes: The table is intentionally wide and uses all 32 cars so that PDF placement and scaling remain visible.
Source: R datasets package.

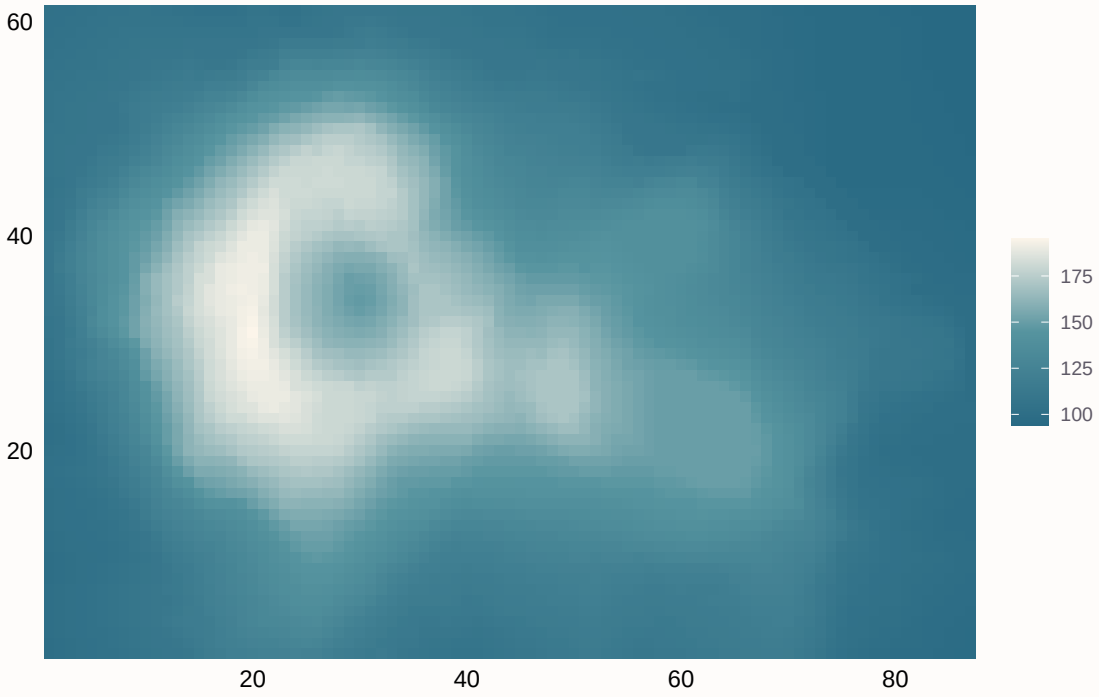
Continuous Scales

Sequential and diverging scales are derived from the primary colours and their tints using OKLab interpolation. Chart 6 and Chart 7 use the same matrix so that the sequential and diverging palettes can be compared directly. This final section is deliberately placed after several other floats so the PDF output has enough material to reveal whether later figures still attach themselves to nearby references.

One last dummy paragraph creates space before the continuous-scale figures. The

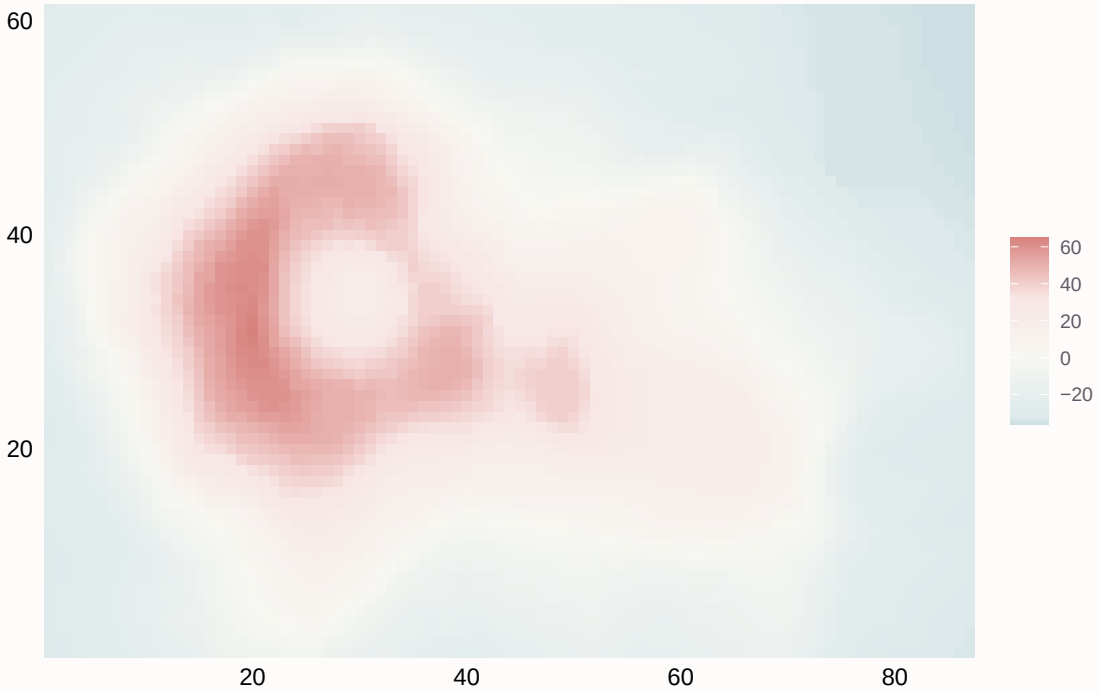
final two charts should still remain above or below nearby text rather than slipping after the references. The barrier before the bibliography is retained only to prevent unresolved floats from entering the reference list.

Chart 6: Sequential colour uses a derived wfmR scale
Blue, teal, and pale yellow encode ordered values



Notes: The palette is sampled with OKLab interpolation.
Source: R datasets package.

Chart 7: Diverging colour centres the substantive zero
Teal and salmon sit on either side of a neutral midpoint



Notes: Use the midpoint argument when zero or another reference value has substantive meaning.
Source: R datasets package.

References

Brewer, Cynthia A., Geoffrey W. Hatchard, and Mark A. Harrower. 2003. "ColorBrewer in Print: A Catalog of Color Schemes for Maps." *Cartography and Geographic Information Science* 30 (1): 5–32. https://doi.org/10.1559/152304003100010929.

Cleveland, William S. 1993. *Visualizing Data*. Hobart Press.

Tufte, Edward R. 2001. *The Visual Display of Quantitative Information*. 2nd ed. Graphics Press.

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